Data Inventory Report

Project Title: Development of Integrated E- Library Management System

Client Organization: Ethiopian Geological Institute (EGI)

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1.1 Introduction

The Ethiopian Geological Institute (EGI) manages large volumes of geological, laboratory, spatial, administrative, and research data from different departments. These datasets support research, mineral exploration, environmental studies, and institutional decision-making. Currently, information is stored in various formats and locations, including digital files, printed documents, scanned reports, and spreadsheets, creating challenges in accessibility, storage, sharing, and long-term preservation.

This Data Inventory Report assesses existing datasets, storage methods, accessibility, ownership, and data management practices within the institution. The findings will support the development of an integrated digital information management system to improve data storage, sharing, accessibility, analytics, and decision-making.

1.2 Objectives of the Data Inventory

The primary purpose of the data inventory assessment is to identify and evaluate all major institutional datasets and information resources that will be incorporated into the proposed digital system.

The specific objectives are:

- To identify all available datasets and information resources across departments.
- To assess the formats, quality, and structure of existing datasets.
- To determine data ownership, accessibility, and usage practices.
- To evaluate existing storage and backup mechanisms.
- To identify gaps and challenges in current data management practices.
- To support the future database architecture and system design process.
- To establish a foundation for metadata standardization and digital transformation.

1.3 Existing Data Sources

The assessment identified several categories of data resources currently used and managed by the institution.

No	Data Source	Description	Format	Owner/Department	Availability
1	Geological Reports	Research findings, geological surveys, and technical reports	PDF/DOCX	Documentation Unit	Available
2	GIS Spatial Data	Geological maps, coordinates, and spatial datasets	SHP/GIS	GIS Department	Available
3	Images and Multimedia	Site photographs, scanned maps, and multimedia files	JPG/PNG/PDF	Archive Unit	Available
4	Administrative Records	User records and operational documents	Database/Excel	ICT Unit	Available

Geological Reports

The organization maintains a large collection of geological reports and technical documents generated through field studies, mineral exploration projects, and research activities. Most of these documents are stored in PDF and Word formats, while some older reports remain available only as printed copies.

GIS Spatial Data

The GIS department manages important spatial datasets including geological maps, geographic coordinates, satellite imagery references, and geospatial layers. These datasets are essential for geological analysis, mapping, and decision support.

Images and Multimedia

The institution also stores large numbers of images, scanned maps, diagrams, and multimedia resources related to geological studies and field operations.

Administrative and Operational Data

Administrative records include staff information, user management data, institutional documents, and operational reports maintained by the ICT and administrative departments.

1.4 Data Classification

The collected information resources can be categorized into structured, semi-structured, and unstructured data.

Structured Data

Structured data refers to organized datasets stored in tables, spreadsheets, or databases.

Examples include:

- User management records
- Metadata records
- Administrative databases
- Inventory records

Semi-Structured Data

Semi-structured data contains partially organized information with flexible formats.

Examples include:

- XML files
- JSON datasets
- Metadata exports
- API response data

Unstructured Data

Unstructured data includes information resources that do not follow a predefined schema.

Examples include:

- PDF reports
- Scanned documents
- Geological maps
- Images and multimedia files

The presence of multiple data types requires the future system to support flexible storage and management mechanisms.

1.5 Current Data Storage Mechanisms

The institution currently uses several storage approaches for managing its information resources.

Local Computer Storage

Many departments store data directly on individual computers used by staff members. While this method provides quick local access, it increases the risk of data loss and duplication.

Shared Network Drives

Some departments use shared network folders for collaborative storage and file sharing. However, the organization of files varies significantly between departments.

External Storage Devices

Portable hard drives and USB devices are commonly used for transferring and backing up files. This practice creates risks related to data security and version control.

Existing Databases

Limited database systems are currently used for administrative operations and selected technical activities. However, these systems are not fully integrated.

1.6 Data Quality Assessment

The assessment identified several data quality issues affecting institutional efficiency and data reliability.

Assessment Area	Observation
Completeness	Some datasets are incomplete or missing records
Consistency	Data formats vary across departments
Accuracy	Manual data entry introduces human errors
Accessibility	Some datasets are difficult to access

Standardization	Lack of metadata and naming standards

1.7 Key Challenges Identified

The assessment identified several operational and technical challenges affecting effective data management.

Fragmented Data Storage

Information resources are distributed across departments, computers, and storage devices without centralized coordination.

Lack of Centralized Database

The organization currently lacks a unified digital repository capable of integrating documents, GIS data, laboratory datasets, and administrative information.

Limited Backup Mechanisms

Backup processes are inconsistent, increasing the risk of accidental data loss.

Inconsistent Metadata Standards

Different departments use different naming conventions and metadata practices.

Difficulty in Data Sharing

Limited interoperability between departments affects collaboration and information exchange.

Limited Web Accessibility

Stakeholders have limited online access to institutional data and information services.

Scalability Challenges

Existing storage approaches may not support future institutional growth and increasing data volumes.

1.8 Recommendations

- Based on the assessment findings, the following recommendations are proposed.

- Develop Centralized Database Architecture
- Establish Metadata Standards
- Digitize Legacy Documents
- Implement Role-Based Access Control
- Introduce Automated Backup and Recovery
- Integrate GIS and Document Management Services
- Improve Data Sharing Mechanisms

1.9 Conclusion

The Data Inventory Assessment confirms the existence of valuable institutional datasets distributed across multiple departments and storage environments. Although important geological, spatial, laboratory, and administrative resources are available, the current management approach faces challenges related to fragmentation, accessibility, standardization, and security.

The findings demonstrate a strong need for an integrated digital platform capable of supporting centralized data ingestion, storage, management, sharing, analytics, and web-based information services. Implementing the proposed system will significantly improve operational efficiency, institutional collaboration, information accessibility, and evidence-based decision-making within the organization.